Resolving the Ambiguity in Bangla Genitive Case: Possessor or Subject?

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Background: In verb-final languages, when the parser reads a sentence incrementally, they utilize case-markers as cues to assign argument roles prior to verb arrival. When a sentence-initial argument is not overtly case-marked, i.e. ambiguous, experimental studies have shown that the parser has a universal preference of assigning it the role of subject/agent [1],[2],[3]. However, this preference has been demonstrated exclusively in contexts involving subject–object ambiguity and unmarked initial arguments. It therefore remains unknown whether subject-first preference persists when (i) the initial argument bears overt but ambiguous case marking, and (ii) the ambiguity extends beyond the subject–object domain.
Aim: Our study addresses these gaps by examining the genitive case in Bangla, which is ambiguous between a possessive modifier (1A) and a subject interpretation (1B). We ask whether the Bangla parser prioritizes subject-first preference or genitive-as-modifier bias when resolving this ambiguity at the first argument.

1A. *[John-er bari] jawa hoye-ni*
[john-GEN house] go happen-not
[pro drop] could not go to John's house today.

1B. *[John-er] [bari] jawa hoye-ni*
[john-GEN] [house] go happen-not
John could not go to the house today.

Our preliminary corpus survey of the Bangla genitive case revealed that genitive-as-modifier uses substantially outnumber genitive-as-subject uses (184 vs. 17), suggesting the possibility of a frequency-based bias toward the former. **Hypotheses:** Against this background, we test three theoretically motivated hypotheses. Hypothesis A or Hyp.A (Structural Prominence): The parser exhibits a subject-first preference, assigning the genitive NP an initial subject interpretation. Hypothesis B or HypB (Frequency Bias): The parser favors the possessive modifier interpretation of the genitive NP, reflecting its higher usage frequency from the corpus evidence. Hypothesis C or Hyp.C (Underspecification): The parser initially underspecifies their interpretation of the genitive NP, showing no online preference during reading.[4]
Experiment Design: (N= 26, Items: 24) A self-paced-reading task (Task 1) was conducted with 1×2 Design, crossing (i) Case Type and (ii) Disambiguation Type. The conditions were: Condition A, subject-disambiguation (S-D) and Condition B, modifier disambiguation (M-D) (2A & B). Bangla correlative constructions were used, where the disambiguation point was at the correlative pronoun (5th position), which refers back to either the genitive marked argument (2nd position) for (S-D) or the accusative marked argument (3rd position) for (M-D).

1st pos	2nd pos	3rd pos	4th pos	5th pos	6th	7th pos

2A. (S-D)
[je meyeta-r]$_k$ [skul-e] [jawa hoye-ni] ***she$_k$*** *khub kan-ch-il-o*
[who.corel girl-GEN] [school-LOC] [go happen-NEG] **that.corel.NOM** very cry-prog-pst.3

The girl who could not go to the school today was crying a lot.
2B. (M-D)
(pro) [je meyeta-r skul-e]$_k$ [jawa hoye-ni] ***shekhan-e$_k$*** *dada pora-e*
(pro-drop) [who.corel girl-GEN school-LOC] [go happen-NEG] **there.corel-LOC** brother teach-pres.3

The girl's school, where [pro-drop] could not go today, is where my brother teaches.

After each sentence, participants answered a binary "Okay/Not Okay" comprehension question (Task 2). This task followed a 2×2 design crossing Question Type (i) Aligned (targeting the same argument as the sentence) vs. Reversed (targeting the alternative argument) and (ii) Disambiguator Type (Subject vs. Modifier). Thus, S-D sentences paired with Subject-Biased questions and M-D with Modifier-Biased questions formed Aligned trials; the opposite pairings formed Reversed trials (See 3i and ii). This arrangement ensured that responses required committing to a specific structural interpretation, and not based on priming due to specific alignment.

3(i). Aligned Question Type

a. (S-D) precedes subject biased question: *meye-ta school jae-ni. thik ki bhul?*
Translation: "The girl did not go to school today". okay or not okay?

b. (M-D) precedes modifier biased question: *keu ekjon meyeta-r school-e jaae-ni, thik ki bhul?*
Translation: "Someone did not go to the girl's school today." okay or not okay?

3(ii). Reversed Question Type

c. (S-D) precedes modifier biased question: *keu ekjon meyeta-r school-e jaae-ni, thik ki bhul?*
Translation: "Someone did not go to the girl's school today." okay or not okay?

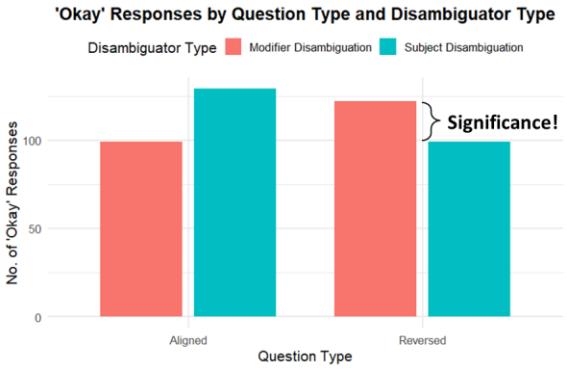
d. (M-D) precedes subject biased question: *meye-ta school jae-ni. thik ki bhul?*
Translation: "The girl did not go to school today". okay or not okay?

Predictions: Hyp.A predicts faster reading times (RT) at region 6 or Spillover, for S-D condition; (S-D < M-D), and more "okay" responses for Subject-Biased questions no matter the condition. Hyp.B predicts faster RT at region 6 for M-D condition (M-D < S-D), and more "okay" responses for Modifier-Biased Questions no matter the condition. Hyp.C predicts no main effects or interaction in RT (S-D = M-D).

Results Participant reading times revealed a numerical trend in which the Subject-disambiguation (S-D) condition was read faster than the Modifier-disambiguation (M-D) condition in the spillover region (6th position); however, this effect did not reach statistical significance ($p = .066$; see Fig. 3.1). In contrast, comprehension responses showed a significant interaction between factors ($\beta \approx -0.44$, $p < .001$; see Fig. 3.2). Under Reversed Question conditions, participants produced significantly more "Okay" responses to subject-biased questions (See 3(ii).d), indicating a subject-first preference when an explicit interpretation was required. Together, these results support Hypothesis A for Bangla: parsers show a subject-first preference, but only under task demands that force an interpretive commitment. Hence, our study extends existing accounts of subject-first preference by showing that, in Bangla, it persists (i) even in the presence of overt case marking, (ii) beyond ambiguity with the object, and (iii) in competition with a more frequent genitive-as-modifier interpretation.

Fig 4.1: RTs across 7 regions (Task 1)

Fig 4.2: Response results (Task 2)



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